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Linking cyberbullying sensitivity to life satisfaction: The mediating role of digital well-being

Vedat Bakır^{1*} and İsmail Kuşci²

Abstract

Background and objective Digital technologies expose individuals to online environments in which perceived risks of cyberbullying may heighten individuals' sensitivity to potential harm, thereby influencing psychological well-being. However, the mechanisms through which cyberbullying sensitivity affects life satisfaction remain unclear. Grounded in Self-Determination Theory, this study examined whether digital well-being mediates the relationship between cyberbullying sensitivity and life satisfaction in a convenience sample of individuals aged 15–39 in Turkey.

Methods A total of 513 individuals (Mage = 22.08, SD = 4.60; range = 15–39; 77.8% female) in Turkey completed validated self-report measures assessing cyberbullying sensitivity, digital well-being, and life satisfaction. Data were analyzed using structural equation modeling with bootstrap procedures (5,000 resamples). Common method bias was assessed using Harman's single-factor test.

Results Cyberbullying sensitivity significantly predicted digital well-being ($\beta = .534, p < .001$), explaining 28.5% of its variance. Digital well-being demonstrated a significant indirect effect in the association between cyberbullying sensitivity and life satisfaction ($\beta = .201, 90\% \text{ CI } [.119, .304], p < .001$). The direct path from cyberbullying sensitivity to life satisfaction was not significant ($\beta = .113, p = .137$), indicating that the association was primarily explained through digital well-being. Harman's single-factor test indicated that common method bias was not a concern (26.69% variance).

Conclusions Digital well-being represents the primary mechanism through which cyberbullying sensitivity is indirectly associated with life satisfaction. Rather than exerting direct effects, online risk perceptions operate through individuals' capacity to maintain psychological balance and self-regulation in digital environments. These findings suggest that interventions should prioritize enhancing digital well-being—through digital literacy, self-regulation skills, and positive online experiences—as a protective factor against the negative consequences of cyberbullying sensitivity. This study advances theoretical understanding by integrating Self-Determination Theory with digital well-being research and provides evidence-based guidance for educators, mental health professionals, and technology designers.

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Keywords Cyberbullying sensitivity, Digital well-being, Life satisfaction, Mediation, Self-determination theory, Convenience sample

Introduction

The rapid expansion of digital technologies and online platforms has improved communication opportunities while simultaneously introducing significant risks for individuals. Among these risks, cyberbullying defined as repeated and intentional harm in online settings has emerged as a particularly serious concern. Empirical evidence indicates that victims frequently report elevated levels of anxiety and depression, alongside diminished self-esteem and reduced life satisfaction [6, 20, 23]. For example, Vieta-Piferrer et al. [36] found that cyberbullying victimization was negatively associated with multiple components of life satisfaction among adolescents, indicating that such experiences can undermine overall subjective well-being. Kara and Aslan [22] also noted that cyber victimization reduces well-being, though this impact may be lessened by self-esteem and social support. These findings indicate that cyberbullying is not only an external risk but also a factor that undermines psychological health and overall quality of life. Importantly, beyond actual exposure to cyberbullying, individuals differ in their level of cyberbullying sensitivity, which refers to heightened cognitive and emotional responsiveness to potential online harassment, even in the absence of direct victimization. Accordingly, cyberbullying sensitivity in the present study refers to perceived vulnerability and anticipatory responses to potential online harassment, rather than to direct experiences of cyberbullying victimization.

Recent research shows that digital well-being is a key factor in understanding the link between cyberbullying sensitivity and life satisfaction. This concept extends beyond time spent online to encompass how individuals utilize digital technologies, their self-control, psychological balance, and satisfaction from digital activities [7]. Mayiwar et al. [30] reported that digital self-control and literacy improve digital well-being, which in turn relates to life satisfaction and social anxiety. Similarly, Chen et al. [9] showed that emotional, cognitive, and behavioral aspects of digital well-being are strongly associated with psychological health. More specifically, network-based analyses indicate that emotional regulation and intrinsic needs satisfaction constitute central mechanisms within the digital well-being system, shaping how digital experiences translate into broader well-being outcomes [9]. Overall, these findings underline that digital well-being shapes how individuals experience the online world and how it affects their life satisfaction. In line with this perspective, recent large-scale evidence indicates that emotional motivations and perceived experiences of social

media use are more strongly associated with life satisfaction than overall usage frequency, highlighting the importance of qualitative aspects of digital engagement [3].

Although earlier studies focusing on cyberbullying victimization have shown adverse effects on well-being, limited research has explored the mediating role of digital well-being in this link. For example, Giumetti and colleagues [17] reviewed evidence showing an inverse link between cyberbullying and well-being, but their work did not explore possible mediating processes. Similarly, Liu et al. [25] observed that cyberbullying lowers life satisfaction, yet the role of digital well-being in this pathway was not considered. Recent empirical findings from university samples further support this pattern, showing a moderate negative association between cyberbullying experiences and life satisfaction, such that higher exposure to online harassment is associated with lower overall life satisfaction [33]. Building on this gap, the current study investigates whether digital well-being fully mediates the link between sensitivity to cyberbullying and life satisfaction. In doing so, the study aims to extend theoretical discussions on both cyberbullying and digital well-being, and to provide practical insights that may guide schools, digital platform designers, and mental health professionals in developing strategies that promote digital well-being.

Theoretical framework

Self-Determination Theory [12] provides a comprehensive framework for understanding how individuals' psychological needs influence their motivation, well-being, and behavioral regulation in various life domains, including digital environments. According to SDT, fulfilling three basic psychological needs—autonomy, competence, and relatedness—is essential for psychological growth, integrity, and well-being, while frustration of these needs is associated with increased aggression and maladaptive behaviors [11].

In the digital context, SDT has been increasingly applied to understand how social media and online platforms can either satisfy or frustrate these basic needs, thereby influencing adolescents' well-being outcomes [4, 37]. Digital well-being, conceptualized within this framework, extends beyond mere screen time to encompass how individuals manage their digital experiences through self-regulation, digital literacy, and satisfaction derived from digital activities [8]. Accordingly, digital well-being can be understood as the functional manifestation of basic psychological need satisfaction in digital contexts [9, 37]. Heightened cyberbullying sensitivity signals a

perceived threat to autonomy, competence, and relatedness in online interactions; rather than directly diminishing global life satisfaction, it disrupts individuals' capacity to maintain psychological balance and self-regulation in digital environments. In particular, threats to relatedness in digital contexts may weaken individuals' sense of social connection and belonging, which is central to digital well-being and subsequent life satisfaction. From an SDT perspective, this positions digital well-being not as a distal correlate, but as the proximal psychological mechanism through which online risk perceptions translate into broader well-being outcomes. When individuals effectively regulate their technology use and maintain psychological balance in digital environments, they are better equipped to internalize motivation and achieve sustained behavioral changes [1].

Cyberbullying sensitivity and psychological vulnerability

This construct is conceptually distinct from cyberbullying victimization in three key ways: (a) it does not require actual exposure to harassment, (b) it reflects anticipatory threat perception rather than experienced harm, and (c) it captures individual differences in vigilance and reactivity that exist independently of victimization history [35]. Individuals higher in this sensitivity tend to perceive potential online harassment more acutely and exhibit stronger emotional reactions when encountering threatening digital content. People with higher sensitivity tend to perceive online harassment more intensely and are more likely to react strongly when victimized. Previous research on cyberbullying victimization shows that experienced online harassment is strongly linked to negative psychological outcomes. Although the present study focuses on sensitivity rather than victimization, research on cyberbullying victimization provides relevant background: victims often report higher depression and anxiety and lower well-being [6, 20, 23]. Contemporary evidence suggests that cyberbullying victimization severely compromises psychological functioning, including decreased happiness, increased anxiety and depression, and disrupted moral cognition [25]. A three-year study with Indian adolescents and young adults found that cyberbullying predicted depression and suicidal thoughts, and these effects continued over time [29]. Research on cyberbullying victimization further suggests that social support can buffer the negative association between cyberbullying experiences and psychological well-being in adolescent populations [21].

Within the SDT framework, cyberbullying sensitivity may threaten basic psychological needs through digital interactions with peers and online communities, which influence individuals' capacity for need satisfaction in digital environments [14]. Individuals with higher cyberbullying sensitivity perceive online harassment more

intensely and experience stronger negative emotional reactions, which subsequently affects their overall psychological functioning [35].

Life satisfaction as an outcome variable

Life satisfaction is regarded as a fundamental dimension of subjective well-being. It reflects how individuals evaluate the overall quality of their lives [13]. Research in positive psychology has shown that life satisfaction is strongly tied to physical health, supportive social ties, and the ability to cope with stress. It is also reasonable to expect that digital well-being contributes to greater life satisfaction, since it offers individuals a sense of balance and control in their technology use and creates more positive online experiences [30]. In addition, meta-analytic studies on social media use suggest that online engagement is linked not only to harmful outcomes but also to certain positive aspects of well-being. However, these associations largely depend on the style and frequency of use as well as on individual psychological processes [28].

Digital well-being as a mediating mechanism

The relationship between online risk perception and life satisfaction is not direct but operates through intermediary psychological processes [37]. We propose that digital well-being serves as a critical mediating mechanism linking cyberbullying sensitivity to life satisfaction through several pathways: First, heightened sensitivity to potential cyberbullying threats threatens the satisfaction of basic psychological needs (autonomy, competence, relatedness), which are fundamental to digital well-being [14]. When individuals perceive high risk of cyberbullying, their sense of autonomy in digital spaces diminishes, competence in managing online interactions decreases, and feelings of relatedness with online communities are disrupted. Second, individuals with higher cyberbullying sensitivity may develop maladaptive digital behaviors—such as excessive monitoring, avoidance, or hypervigilance—that undermine their digital well-being. These disrupted patterns of technology use create a state of psychological imbalance that extends beyond the digital realm to affect overall life evaluation. Third, the temporal dynamics of cyberbullying impact psychological states during the "pericyberbullying period"—the critical time-frame surrounding cyberbullying incidents—suggesting that ongoing sensitivity to such threats maintains a state of reduced well-being [25].

Consistent with SDT, the Relational Model of Cyberbullying [14] emphasizes that immediate social contexts shape individuals' psychological capacities for need satisfaction, which in turn influences their ability to maintain well-being when facing online threats. This perspective acknowledges not only the adverse consequences

of cyberbullying but also the internal mechanisms that shape individuals' digital experiences.

Research questions

Based on self-determination theory and prior empirical findings, the present study addresses the following research questions:

- RQ1. Is cyberbullying sensitivity associated with digital well-being among internet users?
- RQ2. Is digital well-being associated with life satisfaction?
- RQ3. Is cyberbullying sensitivity directly associated with life satisfaction?
- RQ4. Does digital well-being mediate the association between cyberbullying sensitivity and life satisfaction?

Hypotheses

- H1. Cyberbullying sensitivity is associated with digital well-being.
- H2. Digital well-being is associated with life satisfaction.
- H3. Cyberbullying sensitivity is directly associated with life satisfaction.
- H4. Digital well-being mediates the association between cyberbullying sensitivity and life satisfaction.

The hypothesized structural equation model is presented in Fig. 1.

Method

Participants

A total of 513 individuals participated in the study through convenience sampling. The average age was 22.08 years ($SD = 4.60$; range = 15–39). Among them, 77.8% were female ($n = 399$) and 22.2% were male ($n = 114$).

Participants also provided information on their technology use. Regarding social media, 33.7% reported using social media for 3–5 h daily ($n = 173$), 32.4% for 1–3 h ($n = 166$), 25.9% for more than 5 h ($n = 133$), and 8.0% for less than 1 h ($n = 41$). Instagram was the most frequently used platform (42.3%, $n = 217$). With respect to smartphone use, 68.0% reported having used a smartphone for more than five years ($n = 349$), 21.2% for 3–5 years ($n = 109$), 7.4% for 1–3 years ($n = 38$), and 3.3% for less than one year ($n = 17$). As for the purposes of social media use, 78.6% indicated communication and socialization ($n = 403$), 10.3% reported spending time with friends ($n = 53$), and 2.9% reported information seeking ($n = 15$).

Measures

Satisfaction with Life Scale (SWLS)

Diener et al. [13] introduced this measure to evaluate life satisfaction. A Turkish version was prepared by Dağlı and Baysal [10]. The tool includes five statements rated on a 5-point Likert scale and maintains a one-factor model ($\chi^2/df = 1.99$, CFI = 0.99, RMSEA = 0.05). Reported

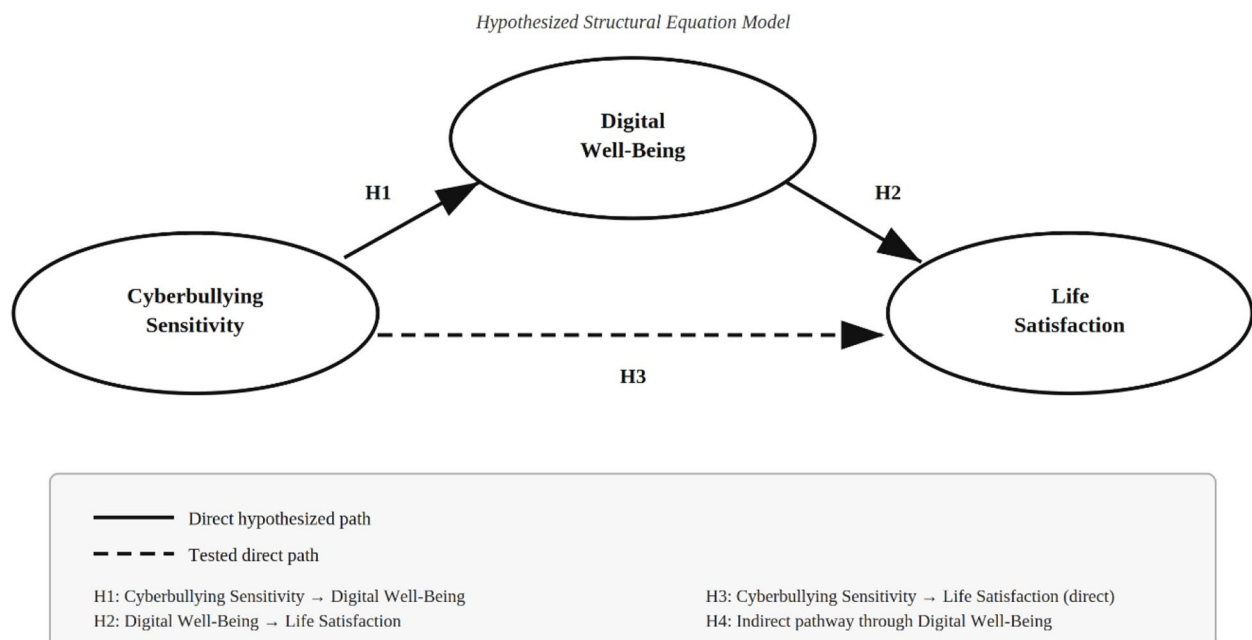


Fig. 1 The hypothesized structural equation model

reliability values were high, with $\alpha = 0.88$ and a test–retest coefficient of 0.97. In the present study, Cronbach's alpha was 0.816. An example item from the Satisfaction with Life Scale is “İdeallerime yakın bir yaşamım vardır” [“In most ways, my life is close to my ideal.”].

Cyberbullying Sensitivity Scale (CSS)

Tanrikulu et al. [35] developed this scale. It measures individuals' awareness and sensitivity to cyberbullying. It contains 14 items in a single-factor structure, rated on a 3-point scale. Exploratory analysis showed that one factor explained 41.2% of the variance, and confirmatory analysis indicated acceptable fit ($\chi^2/df = 2.87$, CFI = 0.90, RMSEA = 0.07). Reported reliability was high, with Cronbach's alpha of 0.90 and split-half reliability of 0.84. In the present study, 13 items were applied, and Cronbach's alpha was 0.905. An example item from the Cyberbullying Sensitivity Scale is “İnternete girdiğimde bilgilerimin başkaları tarafından çalınabileceğini göz önünde tutarım” [“When I go online, I consider the possibility that my personal information may be stolen by others.”].

Digital Well-Being Scale (DWBS)

This scale was created by Arslankara et al. [5]. The instrument consists of 12 items representing three domains: satisfaction in digital contexts, responsible and safe online behavior, and overall digital functioning. Items were rated on a five-point scale, with two items reverse coded. Confirmatory factor analysis indicated an acceptable model fit ($\chi^2/df = 2.24$, CFI = 0.94, RMSEA = 0.06). Reported reliability coefficients for the subscales ranged from 0.71 to 0.78. In this study, the overall Cronbach's alpha was 0.87. An example item from the Digital Well-Being Scale is “Yeni teknolojilere kolay uyum sağlayabilirim” [“I can easily adapt to new technologies.”].

Demographic information form

A demographic questionnaire was developed by the researchers. It was used to gather background information from participants. The form included questions about age, gender, and education level. It also asked about social media habits. Participants reported their average daily time on social media and indicated which platforms they used most often. They were also asked how long they had used a smartphone. Finally, participants described their main purposes for using social media.

Procedure

This study was approved by the Ethics Committee on Scientific Research and Publication Ethics in Social and Human Sciences of Kütahya Dumlupınar University (Meeting No: 2025/09–6, Approval No: 2025–478). The procedures followed the Declaration of Helsinki. Participants were informed about the purpose and ethical

standards of the study prior to participation. For participants aged 15, written parental consent was obtained, and a sample consent form was submitted to the Ethics Committee during the application process. Participants aged 16 and above provided informed consent electronically by selecting the “I agree” option before accessing the questionnaire. Participation was voluntary, and participants were informed that they could withdraw from the study at any time without any consequences. Completing the survey took approximately 10–15 min.

Data were collected using a voluntary participation approach via an online questionnaire created with Google Forms. Participants were reached through online platforms and social media channels using a convenience sampling approach. Additionally, the survey link was shared with individuals who expressed willingness to participate, and further dissemination occurred through personal and family-based WhatsApp groups. No financial or material incentives were provided. Although the survey was initially accessible to a broad age range, responses from participants below the minimum age criterion were excluded prior to analysis. Because the online form required a response to each item before proceeding, no missing data were present in the final dataset. Responses from participants who did not meet the minimum age criterion (15 years) were excluded prior to the analyses, and only eligible cases were retained in the final sample.

Data analysis

Data were analyzed with IBM SPSS 29.0 and AMOS 29.0. Before testing the models, the dataset was examined for normality by inspecting skewness and kurtosis values, and cases identified as extreme outliers were excluded prior to the analyses. Multicollinearity was assessed by examining bivariate correlations among latent variables. Correlation coefficients were below 0.80, indicating that multicollinearity was unlikely to bias the parameter estimates. The measurement structure was confirmed through CFA, and model fit was evaluated using χ^2/df , CFI, TLI, RMSEA, GFI, and RMR [19]. Consistent with commonly accepted guidelines, χ^2/df values below 5, CFI and TLI values close to or above 0.90, and RMSEA values below 0.08 were considered indicative of acceptable model fit. Reliability was verified using Cronbach's α and McDonald's ω . Convergent validity was assessed with AVE and CR values. Discriminant validity was examined using the Fornell–Larcker criterion and the HTMT ratio [15, 18]. Mediation analysis was carried out with SEM, and indirect effects were tested through 5,000 bootstrap resamples [32]. A 90% bootstrap confidence interval was preferred, as recommended in mediation research to provide adequate statistical power and reduce the risk of Type II error in indirect effect testing. This approach

Table 1 Factor loadings and explained variance values of the scales

Scale	Item	Factor Loading	R ²
Satisfaction with Life Scale			
	LS1	.590	.348
	LS2	.689	.475
	LS3	.782	.611
	LS4	.757	.573
	LS5	.634	.402
Cyberbullying Sensitivity Scale			
	CS1	.688	.474
	CS2	.721	.520
	CS3	.469	.220
	CS4	.680	.463
	CS5	.779	.606
	CS6	.699	.489
	CS7	.644	.415
	CS8	.510	.260
	CS9	.649	.421
	CS10	.661	.437
	CS11	.724	.524
	CS12	.555	.308
	CS13	.647	.419
Digital Well-Being Scale			
<i>Digital Satisfaction</i>			
	DWB1	.721	.519
	DWB2	.768	.590
	DWB3	.824	.680
	DWB4	.784	.614
<i>Safe and Responsible Behavior</i>			
	DWB5	.678	.460
	DWB6	.825	.680
	DWB7	.784	.615
	DWB8	.733	.537
<i>Digital Well-Being</i>			
	DWB9	.647	.419
	DWB10	.662	.438
	DWB11	.576	.332
	DWB12	.521	.272

is consistent with prior mediation research suggesting that narrower confidence intervals may be appropriate for indirect effect testing, particularly when the goal is to reduce Type II error and enhance statistical power

[26, 34]. The confidence level was set at 90%, with $p < 0.05$ as the significance criterion. Common method bias was evaluated using Harman's single-factor test [31]. An exploratory factor analysis demonstrated that the first factor explained 26.69% of the variance, which is below the 50% threshold, indicating no serious bias [16].

Results

Measurement model results

The construct validity of the study measures was examined through confirmatory factor analysis (CFA).

The results of the measurement model analysis indicated that the model–data fit was at an acceptable level ($\chi^2/df = 2.737$, RMSEA = 0.058, CFI = 0.902, TLI = 0.891, RMR = 0.061). Factor loadings ranged from 0.469 to 0.825, with 29 out of 30 items exceeding the recommended threshold of 0.50. Given that both the model fit indices and factor loadings were generally within acceptable ranges, structural equation modeling (SEM) analyses were subsequently conducted. The detailed factor loadings and explained variance values are presented in Table 1.

Reliability and validity analyses of the scales

To evaluate the psychometric properties of the instruments used in the study, reliability and validity analyses were conducted.

Internal consistency was examined with Cronbach's alpha and McDonald's omega. The Cyberbullying Sensitivity Scale demonstrated excellent reliability ($\alpha = 0.905$, $\omega = 0.904$). The Digital Well-Being Scale demonstrated good overall reliability ($\alpha = 0.870$, $\omega = 0.865$). At the sub-scale level, Digital Satisfaction ($\alpha = 0.864$, $\omega = 0.864$) and Safe and Responsible Behavior ($\alpha = 0.837$, $\omega = 0.840$) demonstrated acceptable reliability, whereas the Digital Well-Being subdimension fell slightly below the conventional threshold ($\alpha = 0.695$, $\omega = 0.689$). The Satisfaction with Life Scale indicated good reliability ($\alpha = 0.816$, $\omega = 0.819$). The reliability coefficients and validity indices of all scales are summarized in Table 2.

All scales met the composite-reliability cut-off of 0.70, with the Digital Well-Being subdimension being the sole exception. In terms of Average Variance Extracted (AVE),

Table 2 Reliability coefficients and validity indices of the scales

Scale/Factor	α	ω	CR	AVE	MSV	MaxR(H)
Satisfaction with Life	.816	.819	.821	.482	.192	.834
Cyberbullying Sensitivity	.905	.904	.907	.433	.281	.916
Digital Well-Being – Overall	.870	.865	-	-	-	-
Digital Satisfaction	.864	.864	.865	.615	.514	.866
Safe and Responsible Behavior	.837	.840	.842	.573	.514	.852
Digital Well-Being	.695	.689	.695	.365	.402	.704

α Cronbach's alpha, ω McDonald's omega, CR Composite reliability, AVE Average variance extracted, MaxR(H) Maximum construct reliability, MSV Maximum shared variance

Table 3 Fornell–larcker discriminant validity analysis

	1	2	3	4	5
1.Satisfaction with Life	0.694				
2.Cyberbullying Sensitivity	.302***	0.658			
3.Digital Satisfaction	.306***	.386***	0.784		
4.Safe and Responsible Behavior	.439***	.530***	.717***	0.757	
5.Digital Well-Being	.282***	.230***	.634***	.439***	0.604

The diagonal entries indicate the square root of AVE, whereas the values in the lower triangle display the inter-factor correlations

*** $p < .001$

Table 4 HTMT discriminant validity analysis

	1	2	3	4	5
1.Satisfaction with Life					
2.Cyberbullying Sensitivity	.258				
3.Digital Satisfaction	.275	.333			
4.Safe and Responsible Behavior	.374	.453	.628		
5.Digital Well-Being	.208	.188	.487	.339	

HTMT values should be below.85 for the strict criterion and below.90 for the liberal criterion

only Digital Satisfaction (0.615) and Safe and Responsible Behavior (0.573) surpassed the 0.50 criterion. The AVE values for Satisfaction with Life (0.482), Cyberbullying Sensitivity (0.433), and Digital Well-Being (0.365) remained below the threshold. Overall, the psychometric analyses indicated that the instruments used in the study demonstrated generally acceptable reliability and validity properties. However, the relatively low reliability (CR=0.695) and convergent validity (AVE=0.365) of the Digital Well-Being subdimension warrant attention, suggesting that the item structure of this factor may need to be reconsidered. Reliability can also be inferred from CR values [27]; accordingly, the Satisfaction with Life Scale (CR=0.821) and the Cyberbullying Sensitivity Scale (CR=0.907) demonstrated acceptable psychometric adequacy. These reliability and validity findings correspond to the results summarized in Table 2.

Discriminant validity was further examined using the Fornell–Larcker criterion. The inter-factor correlations and square roots of AVE values are presented in Table 3.

To further test discriminant validity, the Heterotrait–Monotrait (HTMT) ratio of correlations was computed. The HTMT results are presented in Table 4.

According to the Fornell–Larcker criterion, all factors except for the Digital Well-Being subdimension met the requirements for discriminant validity. The HTMT analysis further indicated that all values were below the strict cutoff of 0.85, supporting discriminant validity. These results demonstrate that discriminant validity was established across all factors and align with the contemporary criterion proposed by Henseler, Ringle, and Sarstedt [18]. Moreover, the correlations among factors were found to

Table 5 Path coefficients and explained variances of the structural model

Variable/Path	β	S.E	p	R ²
Structural Relationships				
Cyberbullying Sensitivity → Digital Well-Being	.534	.088	<.001	.285
Digital Well-Being → Life Satisfaction	.376	.076	<.001	.199
Cyberbullying Sensitivity → Life Satisfaction (Direct)	.113	.094	.137	
Mediation Effects				
Indirect Effect	.201	.056	<.001	
Total Effect	.314	.059	<.001	
Bootstrap Confidence Intervals (90% CI)				
Indirect Effect Lower Limit (LL)	.119			
Indirect Effect Upper Limit (UL)	.304			

β Standardized regression coefficient, S.E. Standard error, R² Proportion of variance explained

be moderate in magnitude, consistent with theoretical expectations.

Structural mediation model results

Structural equation modeling demonstrated that the proposed mediation model fit the data well. Model indices were $\chi^2/df=2.33$, RMSEA=0.051, CFI=0.948, TLI=0.939, GFI=0.928, and RMR=0.040. These results meet the conventional criteria for a satisfactory model fit [19, 24].

The standardized path coefficients and explained variances for the structural model are summarized in Table 5.

The bootstrap procedure with 5,000 resamples was applied. Cyberbullying sensitivity was significantly associated with digital well-being ($\beta=0.534$, $p<0.001$), and digital well-being significantly predicted life satisfaction ($\beta=0.376$, $p<0.001$). The indirect effect of cyberbullying sensitivity on life satisfaction through digital well-being was significant ($\beta=0.201$, 90% Bootstrap CI [0.119, 0.304], $p<0.001$). The direct effect of cyberbullying sensitivity on life satisfaction was not significant ($\beta=0.113$, $p=0.137$). The final structural equation model with standardized path coefficients is presented in Fig. 2.

The structural equation modeling results indicated that three of the four hypotheses were supported by the data. H3, which examined whether cyberbullying sensitivity was directly associated with life satisfaction, was not supported ($\beta=0.113$, $p=0.137$). This finding suggests that cyberbullying sensitivity, conceptualized as anticipatory vigilance rather than experienced harm, does not exert a direct effect on individuals' global life evaluations when digital well-being is included in the model. H1 examined whether cyberbullying sensitivity was significantly associated with digital well-being. This hypothesis was strongly supported ($\beta=0.534$, $p<0.001$, $R^2=0.285$), indicating that cyberbullying sensitivity was significantly related to digital well-being. H2 examined whether digital well-being

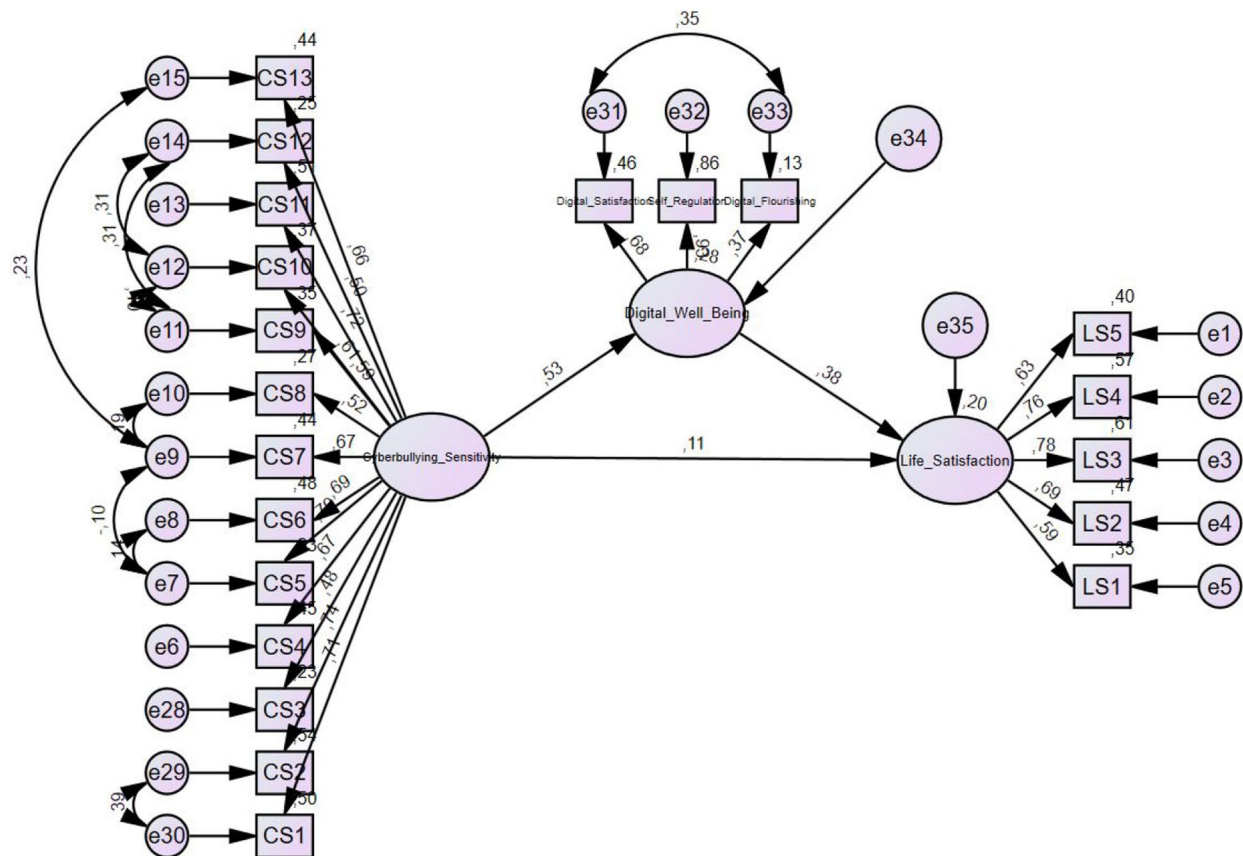


Fig. 2 Structural equation model with standardized path coefficients (β)

was significantly associated with life satisfaction. This hypothesis was also supported ($\beta = 0.376$, $p < 0.001$), demonstrating that higher levels of digital well-being were associated with greater life satisfaction. Finally, H4 examined whether digital well-being statistically mediated the association between cyberbullying sensitivity and life satisfaction. This hypothesis was supported, as evidenced by a statistically significant indirect effect ($\beta = 0.201$, 90% CI [0.119, 0.304], $p < 0.001$). The bootstrap confidence intervals did not include zero, confirming the statistical significance of the mediation pathway. Together, the model explained 19.9% of the variance in life satisfaction ($R^2 = 0.199$).

Discussion and conclusion

The results suggest that digital well-being represents a central psychological mechanism linking cyberbullying sensitivity with life satisfaction. The results further suggest that cyberbullying sensitivity does not show a direct association with life satisfaction; rather, its relationship with life satisfaction emerges indirectly through digital well-being. This pattern indicates that heightened awareness of potential online harm does not automatically translate into a diminished global evaluation of one's life. Instead, the psychological consequences of

cyberbullying sensitivity appear to depend on how individuals regulate, interpret, and manage their digital experiences. Consequently, this research highlights that the direct negative association frequently reported in prior studies (e.g., [20, 23]) can instead be explained through indirect mechanisms such as digital well-being. Notably, much of the existing literature has focused on cyberbullying victimization rather than sensitivity, emphasizing concrete experiences of harm. By contrast, the present findings suggest that when the construct shifts toward perceived vulnerability and awareness, its effects become more indirect and contingent upon internal psychological processes. Importantly, these findings pertain to cyberbullying sensitivity and should not be interpreted as contradicting evidence on the effects of cyberbullying victimization, which involves direct exposure to online harm. However, the absence of a significant direct path should not be interpreted as definitive evidence of full mediation, but rather as a statistical pattern consistent with the proposed indirect pathway.

Cyberbullying sensitivity explained 28.5% of the variance in digital well-being, showing a meaningful link between the two. Vieta-Piferrer et al. [36] likewise found that cyberbullying victimization was negatively associated with multiple components of life satisfaction among

adolescents, indicating that such experiences undermine subjective well-being and may frustrate basic psychological needs. This means that being sensitive to online aggression strongly influences digital well-being. Kara and Aslan [22] further showed that self-esteem and social support can buffer these negative effects. The present findings extend this literature by showing that digital well-being may function as a comparable buffering mechanism.

The mediation model explained 19.9% of the variance in life satisfaction, indicating that digital well-being serves as an important explanatory mechanism in this relationship. This magnitude suggests that digital well-being constitutes a meaningful, though not exclusive, psychological pathway through which online risk perceptions shape subjective well-being. Liu et al. [25] emphasized the detrimental effect of cyberbullying on life satisfaction but did not focus on the mediating mechanisms underlying this relationship. Addressing this gap, the present study demonstrated that digital well-being plays an explanatory role in the link between cyberbullying sensitivity and life satisfaction. The significance of the indirect effect suggests that what matters for individuals' life satisfaction is not merely their sensitivity to cyberbullying but rather how effectively they manage their digital experiences.

Theoretical implications

This study makes notable theoretical contributions to the understanding of cyberbullying and digital well-being. The findings provide empirical support for applying Self-Determination Theory to explain the psychological processes that connect online risk perception with life satisfaction. The mediation model aligns with SDT's proposition that fulfilling basic psychological needs (autonomy, competence, relatedness) reduces negative outcomes, while frustration of these needs intensifies psychological distress [11]. Within this framework, digital well-being functions as the psychological capacity through which individuals preserve or restore autonomy, competence, and relatedness in online environments perceived as risky or threatening. By showing that the association between cyberbullying sensitivity and life satisfaction is primarily explained through digital well-being, we suggest that the psychological implications of sensitivity operate largely via individuals' digital psychological functioning rather than direct pathways.

Second, our results demonstrate how SDT-based processes operate in digital contexts: proximal digital interactions shape individuals' capacity for need satisfaction, which subsequently influences well-being outcomes. This finding aligns with recent relational approaches to cyberbullying research [14]. The significant indirect effect suggests that interventions targeting digital well-being may

effectively buffer the negative consequences of cyberbullying sensitivity. This finding challenges earlier assumptions that cyberbullying's impact on life satisfaction is primarily direct and unchangeable.

Third, the study addresses a critical gap identified by recent scholars. Previous research examining social media use and well-being has produced inconsistent and contradictory findings, partly because studies lacked theoretical frameworks to guide meaningful interpretations [37]. By adopting SDT as our theoretical lens and positioning digital well-being as a mediator, we provide a more nuanced understanding of how online risks translate into life outcomes. Why does cyberbullying sensitivity not directly influence life satisfaction? We propose three interconnected mechanisms:

Cognitive appraisal and coping

Research on the "pericyberbullying period" reveals that psychological changes following cyberbullying are not immediate but evolve through cognitive processing and coping attempts [25]. Individuals high in cyberbullying sensitivity may initially experience heightened vigilance and anxiety in digital spaces. However, whether this sensitivity ultimately affects life satisfaction depends on their capacity to maintain digital well-being through effective self-regulation, digital literacy, and balanced technology use. Those who successfully preserve their digital well-being despite sensitivity to online risks maintain overall life satisfaction.

Need satisfaction dynamics

In SDT, how basic psychological needs are met or blocked in digital settings determines whether online activity supports or harms well-being [11]. Cyberbullying sensitivity threatens need satisfaction in digital environments, but individuals with higher digital well-being possess strategies to restore autonomy (controlling their digital engagement), competence (effectively managing online interactions), and relatedness (maintaining positive online connections). This restoration process explains why the sensitivity-satisfaction link operates through digital well-being.

Behavioral regulation pathways

Self-Determination Theory emphasizes that sustained behavioral change occurs when individuals internalize motivation and develop self-regulatory capacity [1]. Digital well-being reflects individuals' internalized motivation to engage with technology in psychologically healthy ways. Those with poor digital well-being may engage in maladaptive coping (excessive avoidance or problematic overuse), whereas those with strong digital well-being employ adaptive strategies (mindful engagement,

selective exposure, digital detoxing) that prevent cyberbullying sensitivity from eroding life satisfaction.

Our findings indicating that the association operates primarily through digital well-being contrast with some earlier research. Traditional studies found direct links between cyberbullying victimization and reduced life satisfaction, with victims reporting increased depression, anxiety, and suicidal behaviors [6, 20]. However, these studies often measured actual victimization rather than sensitivity, and did not examine intermediate psychological processes. Recent longitudinal research on the pericyberbullying period demonstrates that psychological impacts unfold over time through changes in happiness, moral cognition, and personality traits [25]. Our cross-sectional mediation model complements these findings by identifying digital well-being as the key contemporaneous mechanism. Future longitudinal research could examine whether digital well-being serves as both a mediator and a moderator, potentially buffering the long-term effects of repeated cyberbullying exposure.

Recent reviews examining adolescents' social media engagement from a self-determination perspective suggest that its effects vary according to whether the experience supports or undermines basic psychological needs [37]. Our findings extend this by showing that individual differences in digital well-being—reflecting capacity for need satisfaction in digital contexts—determine whether cyberbullying sensitivity ultimately impacts life satisfaction.

Practical implications

The observed indirect-effect pattern has important practical implications for intervention design. Simply reducing cyberbullying sensitivity through awareness programs or threat detection may be insufficient if individuals lack digital well-being skills. Instead, comprehensive interventions should focus on three levels:

Individual-level interventions

Behavior change technologies designed according to SDT principles can help users internalize motivation for healthy digital behaviors [1]. School counselors and mental health professionals should assess clients' digital well-being alongside cyberbullying experiences. Evidence-based interventions might include digital self-regulation training programs, cognitive restructuring to address maladaptive beliefs about online threats, mindfulness-based approaches to reduce hypervigilance in digital spaces, and skills training in setting digital boundaries and managing screen time. Enhancing individuals' ability to manage their online behavior and strengthen digital literacy can foster greater digital well-being, which subsequently supports overall life satisfaction [30].

Educational interventions

Schools should implement curricula that go beyond traditional "anti-cyberbullying" messaging to teach digital well-being skills. Rather than simply warning students about online risks, educators should help them develop digital literacy to critically evaluate online content and interactions, self-efficacy in managing online conflicts and harassment, strategies for maintaining psychological balance while using technology, and awareness of how digital behaviors affect their overall well-being. Taken together, these results support the close relationship of digital well-being with psychosocial factors such as self-control and resilience, consistent with our proposed mediation model.

Platform-level design

Technology designers can apply SDT principles to create digital environments that support sustained well-being [1]. Social media platforms and online communities could implement features that promote autonomy (user control over exposure, customizable privacy), support competence (effective reporting systems, conflict resolution tools), foster relatedness (positive community-building features, support networks), and provide well-being dashboards that help users monitor and regulate their digital experiences. Taken together, these recommendations translate the study's findings into concrete and actionable strategies that can be directly implemented by practitioners, educators, and platform designers in real-world digital contexts.

Implications for future research

The present findings highlight the need for future research to further examine the mechanisms linking cyberbullying sensitivity, digital well-being, and life satisfaction. Longitudinal and experimental designs are particularly important to clarify the temporal ordering and causal nature of these associations. Additionally, individual difference variables such as personality traits, resilience, and self-regulation capacities should be investigated as potential moderators of the observed mediation pathway. Cross-cultural studies may also help determine whether the role of digital well-being as a protective mechanism generalizes across different sociocultural contexts and age groups. Consistent with this direction, recent research indicates that personality characteristics and self-regulatory capacities shape how digital stressors influence life satisfaction [2], while psychosocial profiles including resilience, loneliness, and self-control play a critical role in cyberbullying-related outcomes [38].

In addition, longitudinal panel designs tracking individuals over extended periods are needed to establish whether changes in digital well-being precede changes

in the cyberbullying sensitivity–life satisfaction relationship. Randomized controlled trials testing digital well-being interventions would further provide causal evidence for the proposed mediation mechanism. Future studies may also examine individual differences in mental health profiles as potential moderators, including personality traits (e.g., resilience, neuroticism), social support quality, offline stress exposure, and digital literacy levels. Finally, multi-level approaches could explore how school climate, family media rules, and peer norms shape individual-level associations among cyberbullying sensitivity, digital well-being, and life satisfaction, while cross-cultural comparisons may determine whether the mediating role of digital well-being operates similarly across socio-cultural contexts.

Limitations and methodological considerations

Despite the contributions of this study, several limitations should be noted. Because the research relied on a cross-sectional design, causal inferences cannot be fully drawn. Although the theoretical model assumes directional links among the variables, other causal paths may also exist. Alternative specifications, such as partial mediation or bidirectional associations, therefore remain plausible and should be tested in future longitudinal studies. Future longitudinal or experimental studies could help clarify temporal order and reduce concerns about reverse causality. For instance, individuals with lower life satisfaction might become more sensitive to cyberbullying and experience reduced digital well-being as a consequence rather than a cause. Accordingly, the findings should be interpreted as correlational rather than causal, and the proposed mediation model reflects theoretically grounded associations consistent with the data rather than evidence of causal effects. Therefore, the present findings do not claim full mediation in a causal sense, but rather indicate that digital well-being statistically accounts for the observed association within the tested model.

Although alternative model specifications (e.g., partial mediation or reversed causal ordering) are theoretically plausible, the present study focused on the model derived from Self-Determination Theory. Testing competing models was beyond the scope of the current analysis and would require longitudinal or experimental designs to establish temporal precedence. Future research should explicitly compare alternative model specifications to further evaluate the robustness of the proposed mediation pathway.

Another limitation involves the exclusive use of self-report measures, which may introduce response bias. Although the results of Harman's single-factor test (26.69%) indicated that common method bias was not problematic, future research could include behavioral indicators such as screen-time records or platform

activity logs. Using ecological momentary assessment or retrospective social media data may also provide a more dynamic understanding of how cyberbullying relates to psychological adjustment [25].

The convenience sampling strategy limits generalizability, as participants were recruited online through digital platforms and social media channels, resulting in a heterogeneous age and education profile (15–39 years). Cyberbullying affects individuals across age groups, with different developmental stages presenting unique vulnerabilities [6]. Younger adolescents may lack the cognitive resources for effective digital self-regulation, while older adults may face different forms of online harassment (e.g., in workplace contexts). Cultural factors also shape how individuals interpret cyberbullying and maintain digital well-being, suggesting caution in generalizing findings beyond Turkish internet users. An additional limitation concerns the psychometric properties of the cyberbullying sensitivity scale. Although the overall measurement model demonstrated acceptable fit, one subdimension yielded AVE and composite reliability values below commonly recommended thresholds. While such values do not necessarily invalidate the construct, they suggest that this subdimension may not fully capture the intended latent variance. Accordingly, the mediation findings involving cyberbullying sensitivity should be interpreted with appropriate caution. Future research should further refine this subdimension and re-examine its measurement properties using alternative indicators and independent samples.

The Digital Well-Being Scale's psychometric properties, particularly the relatively low composite reliability for one subdimension ($CR = 0.695$), suggest that this construct requires further refinement and validation. Future research should develop more robust measures capturing the multidimensional nature of digital well-being with stronger discriminant validity. These limitations directly inform directions for future research, underscoring the need for longitudinal and experimental designs, multi-method assessments, and more diverse samples to more precisely capture the dynamic and contextual nature of cyberbullying sensitivity and digital well-being. Although AVE values slightly below 0.50 are sometimes considered acceptable in applied SEM research when composite reliability is adequate and the construct is theoretically grounded, these values nonetheless represent a limitation when interpreting the convergent validity of the digital well-being construct in the present study.

Conclusion

This study demonstrates that digital well-being plays a key mediating role in the association between cyberbullying sensitivity and life satisfaction. The lack of a direct effect, alongside a strong indirect path, suggests that

individuals maintain psychological balance and self-regulation in digital settings. This mechanism appears to be the main route through which perceptions of online risk influence overall life satisfaction. These findings provide both theoretical advancements in understanding cyberbullying's psychological impacts and practical guidance for developing effective interventions. Rather than focusing solely on reducing cyberbullying exposure, mental health professionals, educators, and technology designers should prioritize fostering digital well-being as a protective factor that enables individuals to navigate online spaces while maintaining life satisfaction.

Abbreviations

CSS	Cyberbullying Sensitivity Scale
DWBS	Digital Well-Being Scale
SWLS	Satisfaction with Life Scale
SEM	Structural Equation Modeling
SDT	Self-Determination Theory

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Not applicable.

AI declaration

The authors used Grammarly Pro to improve readability and grammatical clarity with caution.

Code availability

Not applicable.

Authors' contributions

The study idea and design were developed collaboratively by both authors. Data collection was carried out by both authors. VB led the literature review and the writing of the discussion section. IK played a primary role in conducting the statistical analyses and interpreting the findings. Both authors critically reviewed the entire manuscript, made substantial revisions, and read and approved the final version.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ethical restrictions but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and was approved by the Ethics Committee on Scientific Research and Publication Ethics in Social and Human Sciences of Kütahya Dumlupınar University (Meeting No: 2025/09–6, Approval No: 2025–478). Written informed consent was obtained from all participants prior to data collection. For participants under the age of 16, written informed consent to participate was additionally obtained from their parents or legal guardians.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Alberts L, Lyngs U, Lukoff K. (2024). Designing for sustained motivation: a review of self-determination theory in behaviour change technologies. *Interact Comput.* <https://doi.org/10.48550/arXiv.2402.00121>
2. Alphenaar LEV, Shiner RL, Chavez Arana C, Prinzie P. Social media and subjective well-being: the moderating role of personality traits. *J Happiness Stud.* 2025;26:61. <https://doi.org/10.1007/s10902-025-00898-0>.
3. Andersen AIO, Skogen JC, Hjetland GJ, Finserås TR, Bøe T, Colman I, et al. The relationship between different aspects of social media use and mental health problems and life satisfaction among Norwegian students. *Computers in Human Behavior Reports.* 2026;21:100881. <https://doi.org/10.1016/j.chbr.2025.100881>.
4. Arpacı I, Kuscı I. A hybrid SEM-ANN approach for predicting the impact of psychological needs on satisfaction with generative AI use. *Technol Knowl Learn.* 2025;30(4):2329–50. <https://doi.org/10.1007/s10758-025-09817-x>.
5. Arslankara VB, Demir A, Öztaş Ö, Usta E. Digital well-being scale validity and reliability study. *J Teach Educ Lifelong Learn.* 2022;4(2):263–74. <https://doi.org/10.51535/tell.1206193>.
6. Bansal S, Garg N, Singh J, Van Der Walt F. Cyberbullying and mental health: past, present and future. *Front Psychol.* 2024;14:1279234. <https://doi.org/10.3389/fpsyg.2023.1279234>.
7. Bhattacharya S, Bhattacharya S, Vallabh V, Marzo RR, Juyal R, Gokdemir Ö. Digital well-being through the use of technology – a perspective. *Int J MCH AIDS.* 2023;12(1):e588. <https://doi.org/10.21106/ijma.588>.
8. Büchi M. Digital well-being: in search of a definition. *New Media Soc.* 2024;26(1):35–52. <https://doi.org/10.1177/1461448211056851>.
9. Chen S, Ebrahimi OV, Cheng C. New perspective on digital well-being by distinguishing digital competency from dependency: a network approach. *J Med Internet Res.* 2025;27:e70483. <https://doi.org/10.2196/70483>.
10. Dağlı A, Baysal N. Yaşam Doyumu Ölçeği'nin Türkçe'ye uyarlanması: Geçerlik ve güvenilirlik çalışması. *Elektronik Sosyal Bilimler Dergisi.* 2016;15(59):1250–62. <https://doi.org/10.17755/esosder.75955>.
11. Deci EL, Ryan RM. (2012). Motivation, personality, and development within embedded social contexts: an overview of self-determination theory. In R. M. Ryan (Ed.), *The Oxford handbook of human motivation* (pp. 85–107). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780195399820.013.0006>
12. Deci EL, Ryan RM. *Intrinsic motivation and self-determination in human behavior.* New York: Springer Science & Business Media; 2013.
13. Diener E, Emmons RA, Larsen RJ, Griffin S. The satisfaction with life scale. *J Pers Assess.* 1985;49(1):71–5. https://doi.org/10.1207/s15327752jpa4901_13.
14. Doty JL, Barlett CP, Gabrielli J, Yourell JL, Su YW, Waasdorp TE. A model of personal relationships and cyberbullying perpetration among adolescents: a person environment model. *J Soc Pers Relat.* 2024;41(7):1980–2005. <https://doi.org/10.1177/02654075241227394>.
15. Fornell C, Larcker DF. Evaluating structural equation models with unobservable variables and measurement error. *J Mark Res.* 1981;18(1):39–50. <https://doi.org/10.2307/3151312>.
16. Fuller CM, Simmering MJ, Atinc G, Atinc Y, Babin BJ. Common methods variance detection in business research. *J Bus Res.* 2016;69(8):3192–8. <https://doi.org/10.1016/j.jbusres.2015.12.008>.
17. Giumetti GW, Kowalski RM. Cyberbullying via social media and well-being. *Curr Opin Psychol.* 2022;45:101314. <https://doi.org/10.1016/j.copsyc.2022.101314>.
18. Henseler J, Ringle CM, Sarstedt M. A new criterion for assessing discriminant validity in variance-based structural equation modeling. *J Acad Mark Sci.* 2015;43(1):115–35. <https://doi.org/10.1007/s11747-014-0403-8>.
19. Hu L-t, Bentler PM. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct Equ Modeling.* 1999;6(1):1–55. <https://doi.org/10.1080/10705519909540118>.
20. Hu Y, Bai Y, Pan Y, Li S. Cyberbullying victimization and depression among adolescents: a meta-analysis. *Psychiatr Res.* 2021;305:114198. <https://doi.org/10.1016/j.psychres.2021.114198>.
21. Kahi M, Heidari Z, Esmaili M, Samouei R, Amini-Rarani M. From screens to minds: the connection between cyberbullying, psychological well-being, and

- the mediating role of perceived social support in high school students. *Adv Public Health*. 2024;2024:6662382. <https://doi.org/10.1155/2024/6662382>.
22. Kara E, Aslan H. Cyber victimization and subjective well-being: protective roles of self-esteem and social support among Turkish gifted youths. *BMC Psychol*. 2025;13:341. <https://doi.org/10.1186/s40359-025-02636-w>.
 23. Kim S, Colwell SR, Kata A, Boyle MH, Georgiades K. Cyberbullying victimization and adolescent mental health: evidence of differential effects by sex and mental health problem type. *J Youth Adolesc*. 2018;47(3):661–72. <https://doi.org/10.1007/s10964-017-0678-4>.
 24. Kline RB. Principles and practice of structural equation modeling. New York: Guilford publications; 2023.
 25. Liu X, Liu M, Kang X, Han N, Liao Y, Ren Z. More cyberbullying, less happiness, and more injustice—psychological changes during the pericyberbullying period: quantitative study based on social media data. *J Med Internet Res*. 2025;27:e64451. <https://doi.org/10.2196/64451>.
 26. MacKinnon DP, Lockwood CM, Williams J. Confidence limits for the indirect effect: distribution of the product and resampling methods. *Multivar Behav Res*. 2004;39(1):99–128. https://doi.org/10.1207/s15327906mbr3901_4.
 27. Malhotra NK, Dash S. Marketing Research: An Applied Orientation. 6th Edition. Delhi: Pearson-Dorling Kindersley; 2011. p. 126–33.
 28. Marciano L, Lin J, Sato T, Saboor S, Viswanath K. Does social media use make us happy? A meta-analysis on social media and positive well-being outcomes. *SSM – Mental Health*. 2024;4:100331. <https://doi.org/10.1016/j.jssm.mh.2024.100331>.
 29. Maurya C, Muhammad T, Dhillon P, Maurya P. The effects of cyberbullying victimization on depression and suicidal ideation among adolescents and young adults: a three-year cohort study from India. *BMC Psychiatry*. 2022;22:599. <https://doi.org/10.1186/s12888-022-04238-x>.
 30. Mayiwar L, Asutay E, Tinghög G, Väst D, Barrafrém K. Determinants of digital well-being: digital self-control, literacy, and information ignorance as predictors of digital well-being, life satisfaction and social anxiety. *AI & Soc*. 2025. <https://doi.org/10.1007/s00146-024-02071-2>.
 31. Podsakoff PM, MacKenzie SB, Lee J-Y, Podsakoff NP. Common method biases in behavioral research: a critical review of the literature and recommended remedies. *J Appl Psychol*. 2003;88(5):879–903. <https://doi.org/10.1037/0021-9010.88.5.879>.
 32. Preacher KJ, Hayes AF. Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behav Res Methods*. 2008;40(3):879–91. <https://doi.org/10.3758/brm.40.3.879>.
 33. Radjaa Z, Khadidja M. Cyberbullying and life satisfaction among university students. *Sci Educ Innov Context Modern Probl*. 2025;8(11):1083–8. <https://doi.org/10.56334/sei/8.11.88>.
 34. Shrout PE, Bolger N. Mediation in experimental and nonexperimental studies: new procedures and recommendations. *Psychol Methods*. 2002;7(4):422–45. <https://doi.org/10.1037/1082-989X.7.4.422>.
 35. Tanrıkulu T, Kınay H, Arıcak OT. Siber Zorbalığa İlişkin Duyarlılık Ölçeği: Geçerlik ve güvenirlik çalışması. *Trakya Univ J Educ*. 2013;3(1):38–47.
 36. Vieta-Piferrer J, Oriol X, Miranda R. Longitudinal associations between cyberbullying victimization and cognitive and affective components of subjective well-being in adolescents: a network analysis. *Appl Res Qual Life*. 2024;19(6):2967–89. <https://doi.org/10.1007/s11482-024-10363-4>.
 37. West M, Rice S, Vella-Brodrick D. Adolescent social media use through a self-determination theory lens: a systematic scoping review. *Int J Environ Res Public Health*. 2024;21(7):862. <https://doi.org/10.3390/ijerph21070862>.
 38. Yazıcı-Kabadayı T, Mercan N, Öztemel E. Cyberbullying roles and psychosocial dynamics: a latent profile analysis of loneliness, resilience, and self-regulation in adolescents. *BMC Public Health*. 2025;25:1478. <https://doi.org/10.1186/s12889-025-22745-w>.

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